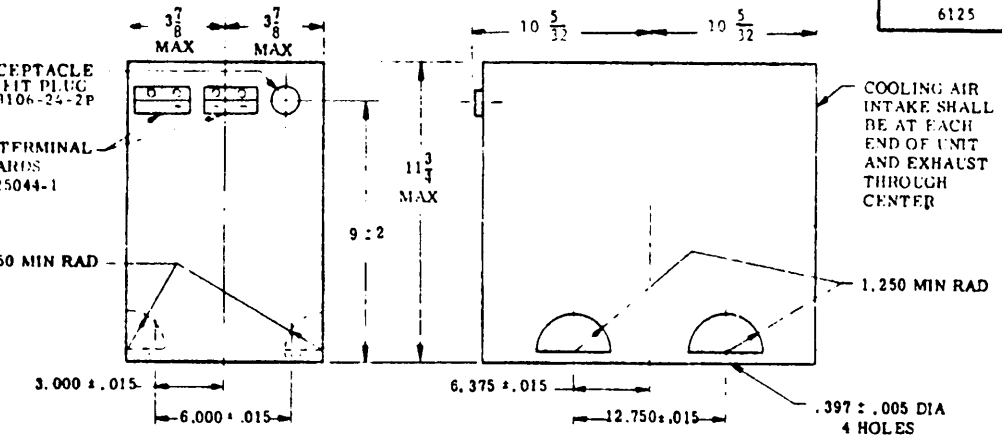
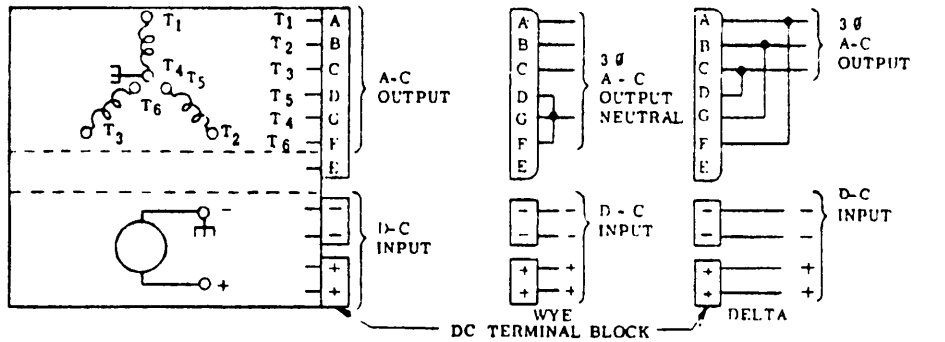




CONNECTIONS EXTERNAL TO THE INVERTER NECESSARY FOR WYE (115/200 VOLT) AND DELTA (115 VOLT) CONNECTIONS. FOR 1 Ø CONNECTION, SEE NOTE 1.

MS PART NO.
MS25202-1

PARAMETERS AND CONDITIONS		3 Ø	1 Ø
INPUT:	VOLTS, DC	28	28
OUTPUT:	VOLTS, AC	115/200	115
	FREQUENCY, HZ	400	400
	CLASS A OPERATION:		
	VA TO 35,000 FEET ALTITUDE AND -10° C TEMPERATURE	5000	5000
	CLASS B OPERATION:		
	VA TO 50,000 FEET ALTITUDE AND 20° C TEMPERATURE	3500	3500
	VA TO 65,000 FEET ALTITUDE AND 20° C TEMPERATURE	1750	1750
	PF MINIMUM LAG, PERCENT	90	90
	PF MINIMUM LEAD, PERCENT	95	95
	MINIMUM OUTPUT VOLTAGE AT 200 PERCENT RATED CURRENT	50 85	50
MINIMUM EFFICIENCY, PERCENT AT RATED LOAD, SEA LEVEL, AND 25° ± 15° C		60	55
WEIGHT:	POUNDS, MAXIMUM	76	76

(B) ENTIRE STANDARD REVISED AND REDRAWN

APPROVED 16 AUG 56 REVISED (A) 14 APRIL 72 (B) 27 September 1976

This military standard is approved by NAVAL AIR SYSTEMS COMMAND, Department of the Navy and is mandatory for use by the Naval Activity. All other military activities are required to employ this standard where suitable.

P.A. NAVY - AS Other Cust	TITLE INVERTER: 5000VA, 3/1 PHASE, CLASS A 7500VA, 3/1 PHASE, CLASS B	MILITARY STANDARD MS25202(AS)
PROCUREMENT SPECIFICATION MIL-I-7032	SUPERSEDES	SHEET 1 OF 1

DD FORM 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE

PLATE NO 73071

PROJECT NO. 6125-A142

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:05Z
Check the source to verify that this is the current version before use.

FOR ANY COMBINATION OF CONDITIONS LISTED BELOW, THE OUTPUT VOLTAGE AND FREQUENCY SHALL REMAIN WITHIN THE INDICATED LIMITS.

OPERATING CONDITIONS	OUTPUT LIMITS	
	VOLTAGE	FREQUENCY HZ
INPUT VOLTAGE VARIED BETWEEN 26 AND 29 VOLTS LOAD VARIED FROM 0 TO FULL LOAD AMBIENT TEMPERATURE CLASS A OPERATION ALTITUDE VARIED WITHIN THE RANGE ESTABLISHED BY CURVE II OF MS33543. THE UNIT SHALL DELIVER RATED OUTPUT TO 35,000 FEET ALTITUDE AND -10°C TEMPERATURE. CLASS B OPERATION ALTITUDE VARIED WITHIN THE RANGE ESTABLISHED BY CURVE 1 OF MS33543, EXCEPT THAT THE AMBIENT TEMPERATURE FOR THIS UNIT SHALL NEVER EXCEED 85°C. THE UNIT SHALL DELIVER RATED OUTPUT TO 50,000 FEET ALTITUDE AND 50 PERCENT RATED OUTPUT TO 65,000 FEET ALTITUDE, BOTH AT 20°C TEMPERATURE. LOAD POWER FACTOR VARIED FROM .90 LAG TO .95 LEAD.	LOW 110/192 HIGH 120/208	LOW 390 HIGH 410

REQUIREMENTS:

1. THIS INVERTER DESIGN SHALL BE SUCH THAT IT MAY BE LOADED EITHER 3 Ø OR 1 Ø BUT NOT SIMULTANEOUSLY. WHEN USED AS A SINGLE-PHASE INVERTER, THE UNIT SHALL BE CONNECTED IN DELTA AND THE LOAD CONNECTED TO THE TERMINALS A(T_1) AND B(T_2). FOR 3 Ø AIRCRAFT APPLICATIONS, THE LOADS PER PHASE SHALL NOT DIFFER BY MORE THAN 10 PERCENT OF THE INVERTER RATING.

2. THE AIRCRAFT MANUFACTURER SHALL ALLOW AT LEAST ONE INCH CLEARANCE IN ADDITION TO THE MAXIMUM DIMENSIONS SHOWN ON THIS DRAWING FOR INSTALLATION AND VENTILATION.

3. THE WEIGHT AND DIMENSIONS SHOWN ON THIS DRAWING SHALL INCLUDE THE WEIGHT AND DIMENSIONS OF ALL AUXILIARY APPARATUS NECESSARY TO MAKE THE UNIT CONFORM TO THE PROCUREMENT SPECIFICATION.

4. A MEANS OF ADJUSTING THE VOLTAGE SHALL BE ON THE CONNECTOR END AND ABOVE THE HORIZONTAL CENTERLINE OF THE INVERTER.

5. PERMISSION OF THE PROCURING ACTIVITY SHALL BE OBTAINED FOR APPLICATION OF THIS UNIT OUTSIDE OF THE SPECIFIED LIMITS.

6. TWO PLACARDED PIN JACKS SHALL BE PROVIDED ON THE CONNECTOR END AND ABOVE THE HORIZONTAL CENTERLINE OF THE UNIT BETWEEN WHICH THE VOLTAGE BETWEEN A(T_1) AND G(T_4) CAN BE MEASURED. THESE PIN JACKS SHALL ACCEPT A PROD PIN .078 ±.002 INCHES IN DIAMETER.

7. RADIO INTERFERENCE - THE INVERTER SHALL MEET THE REQUIREMENTS OF MIL-I-6181 WITH THE FOLLOWING RELAXATIONS:

   A. BROADBAND RADIATED INTERFERENCE LIMITS

      15 DB BETWEEN 0.15 MHZ AND 0.5 MHZ
      3 DB BETWEEN 0.5 MHZ AND 1.5 MHZ
      2 DB BETWEEN 35 MHZ AND 90 MHZ

   B. BROADBAND CONDUCTED INTERFERENCE LIMITS

      10 DB BETWEEN 0.2 MHZ AND 0.9 MHZ

8. VIBRATION - THE INVERTER SHALL BE SUBJECTED TO DURABILITY AND PERFORMANCE TESTS AS FOLLOWS:

   A. DURABILITY - THE INVERTER SHALL BE OPERATED AT FULL LOAD DURING THIS TEST. THE INVERTER SHALL BE SUBJECTED TO HARMONIC OR CIRCULAR MOTION APPLIED TO THE MOUNTING BASE. THE AMPLITUDE SHALL BE 0.015 INCH (MAXIMUM TOTAL EXCURSION OF 0.03 INCH). THE FREQUENCY OF THE VIBRATION SHALL BE ESTABLISHED BY CYCLIC OPERATION FROM APPROXIMATELY 10 CPS TO 55 CPS AND BACK TO 10 CPS. THE DURATION OF ONE SUCH CYCLIC OPERATION SHALL BE BETWEEN 1 TO 5 MINUTES. THE INVERTER SHALL BE VIBRATED CONTINUOUSLY FOR 2 HOURS IN THE DIRECTION OF EACH OF THE 3 MAJOR AXES. AT THE END OF THE TEST PERIOD, THE INVERTER SHALL BE THOROUGHLY INSPECTED FOR DAMAGE OR OTHER DEFECTS RESULTING FROM THE VIBRATION TEST.

   B. PERFORMANCE - THE INVERTER SHALL BE SUBJECTED TO SIMPLE HARMONIC OR CIRCULAR MOTION APPLIED TO THE MOUNTING BASE. THE AMPLITUDE SHALL BE 0.015 INCH (MAXIMUM TOTAL EXCURSION OF 0.03 INCH) UP TO 60 CPS, THE ACCELERATION SHALL BE (ABOVE 60 CPS) LIMITED TO 5G. THE INVERTER SHALL BE VIBRATED OVER THE FREQUENCY RANGE OF 20 TO 500 CYCLES IN 15 MINUTES ALONG EACH OF ITS THREE MAJOR AXES. THE UNIT SHALL SHOW NO SIGNS OF INSTABILITY OR HARMFUL ARCING AND THE OUTPUT VOLTAGE MODULATION SHALL NOT EXCEED 2.0 PERCENT. FOLLOWING THIS TEST, THE INVERTER SHALL PASS THE TEST SPECIFIED IN PARA. 4.6.4(a) OF MIL-I-7032F.

APPROVED 16 AUG 56 REVISED (B) FOR CHANGES SEE SHEET 1

This military standard is approved by BUAER and BUSHIPS and SYSTEMS COMMAND, Department of the Navy and shall be used by the activity. All other military services are required to comply the standard when invited.

P.A. NAVY - AS Other Cust	TITLE INVERTER; 5000VA, 3/1 PHASE, CLASS A 3500VA, 3/1 PHASE, CLASS B	MILITARY STANDARD MS25202(AS)
PROCUREMENT SPECIFICATION MIL-I-7032	SUPERSEDES	SHEET 2 OF 3

Source: https://assist.dla.mil -- Downloaded: 202640-23T15:05Z
Check the source to verify that this is the current version before use.

9. THE INVERTER SHALL MEET ALL REQUIREMENTS WITHOUT ANY WARMUP.

10. THE INVERTER SHALL BE PROVIDED WITH A METER, TIME TOTALIZER CONFORMING TO MS17322-10. THE METER SHALL BE LOCATED ON THE CONNECTOR END AND ABOVE THE CENTERLINE OF THE INVERTER.

11. THE INVERTER SHALL BE CAPABLE OF SPECIFIED PERFORMANCE WHEN OPERATED FROM A DC POWER SOURCE CAPABLE OF MAINTAINING A DC VOLTAGE TO THE INVERTER WHICH WILL BE AT ITS MINIMUM STEADY-STATE VALUE WITHIN 200 MILLISECONDS FROM APPLICATION.

12. THE INVERTER SHALL BE PROVIDED WITH A STATIC SEMI-CONDUCTOR VOLTAGE AND FREQUENCY REGULATOR.

NOTES:

1. DIMENSIONS IN INCHES.

2. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS, OR REQUEST FOR PROPOSAL EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.

3. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

APPROVED 16 AUG 56 REVISED (B) FOR CHANGES SEE SHEET 1

This military standard is approved by NAVAL AIR SYSTEMS COMMAND, Department of the Navy and is mandatory for use by the Department of the Navy. All other military activities are required to employ this standard where suitable.

P.A. NAVY - AS Other Cust	TITLE INVERTER; 5000VA, 3/1 PHASE, CLASS A 3500VA, 3/1 PHASE, CLASS B	MILITARY STANDARD MS25202(AS)
PROCUREMENT SPECIFICATION MIL-I-7032	SUPERSEDES	SHEET 1 OF 1

DD FORM (MAR 72) 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE PLATE NO 23071

Source: https://assist.dla.mil -- Downloaded: 202610-23T15:05Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:05Z
Check the source to verify that this is the current version before use.